L3 informatique

Course Notes
Software Engineering

Course 8

Introduction to development methods

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Model and methods

- Software development process a set of interrelated and interconnected activities in which user needs are transformed into software
  - A process describes 2 important things:
    - Activities (steps) (WHAT?)
    - The sequence of activities (WHEN?)
- Software life cycle models.
  - Models characterize the links and relationships between the different stages of the software life cycle.
- Software life cycle methods.
  - The methods make it possible to implement software development according to a model by organizing the different stages of the software life cycle.
- Method = Approach + Language

Type of development process

- Predictive processes (Ex: V, Cascade, Spiral)
  - Very precise and detailed planning.
  - The development team completes the development phases in a strict order.
  - Rigorous planning does not allow for changes in user needs and causes members of the development team to work slowly.
- Adaptive processes (Ex: UP: RUP, 2TUP,… and agile: Scrum, XP….)
  - They correspond to an iterative life cycle which considers changes in user needs and technical architecture.
  - They favor delivery in installments starting with the functionalities useful to the customer.

Adaptive processes

- They correspond to an iterative and incremental life cycle They favor delivery in installments starting with the functionalities useful to the customer.
- There are 3 main families of development processes:
  - Unify process (UP): RUP, 2TUP…….
  - Agile method: XP, Scrum, RAD…….
  - Other method: AUP, XUP, EssUP,……..

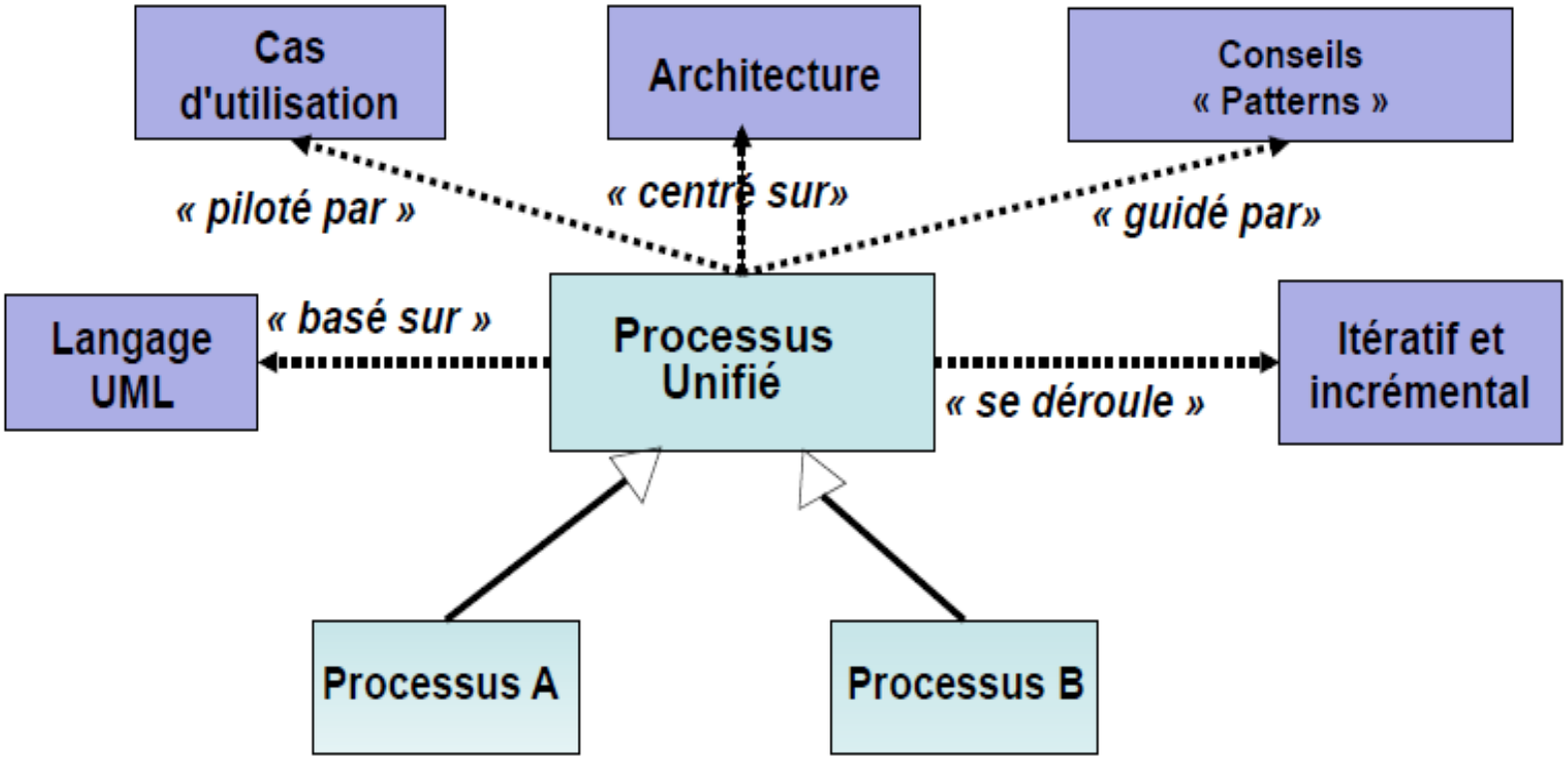
Iteration and incremental

- iterative development: complete system in each iteration
- incremental development: partial system gradually completed at each iteration

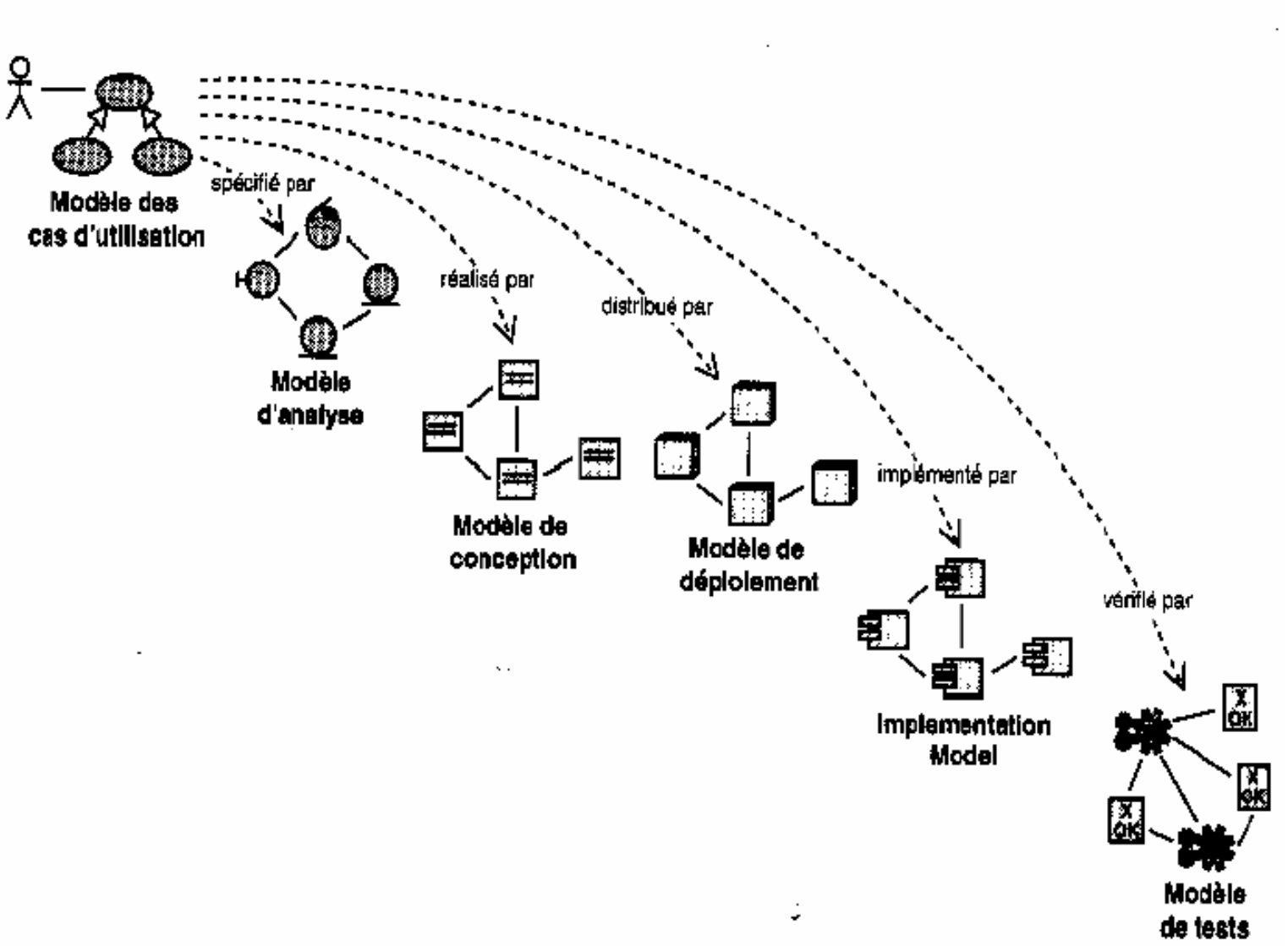
Method

- Method (development context)
  - A reproducible approach to obtain a reliable result.
- Information system development method
  - The rules for modeling and building software systems in a reliable and reproducible manner the implementation
  - rules which describe the sequence of actions, ordering of tasks and the distribution of tasks

Processus unifier (UP)



Driven by use cases



Centered on architecture and pattern

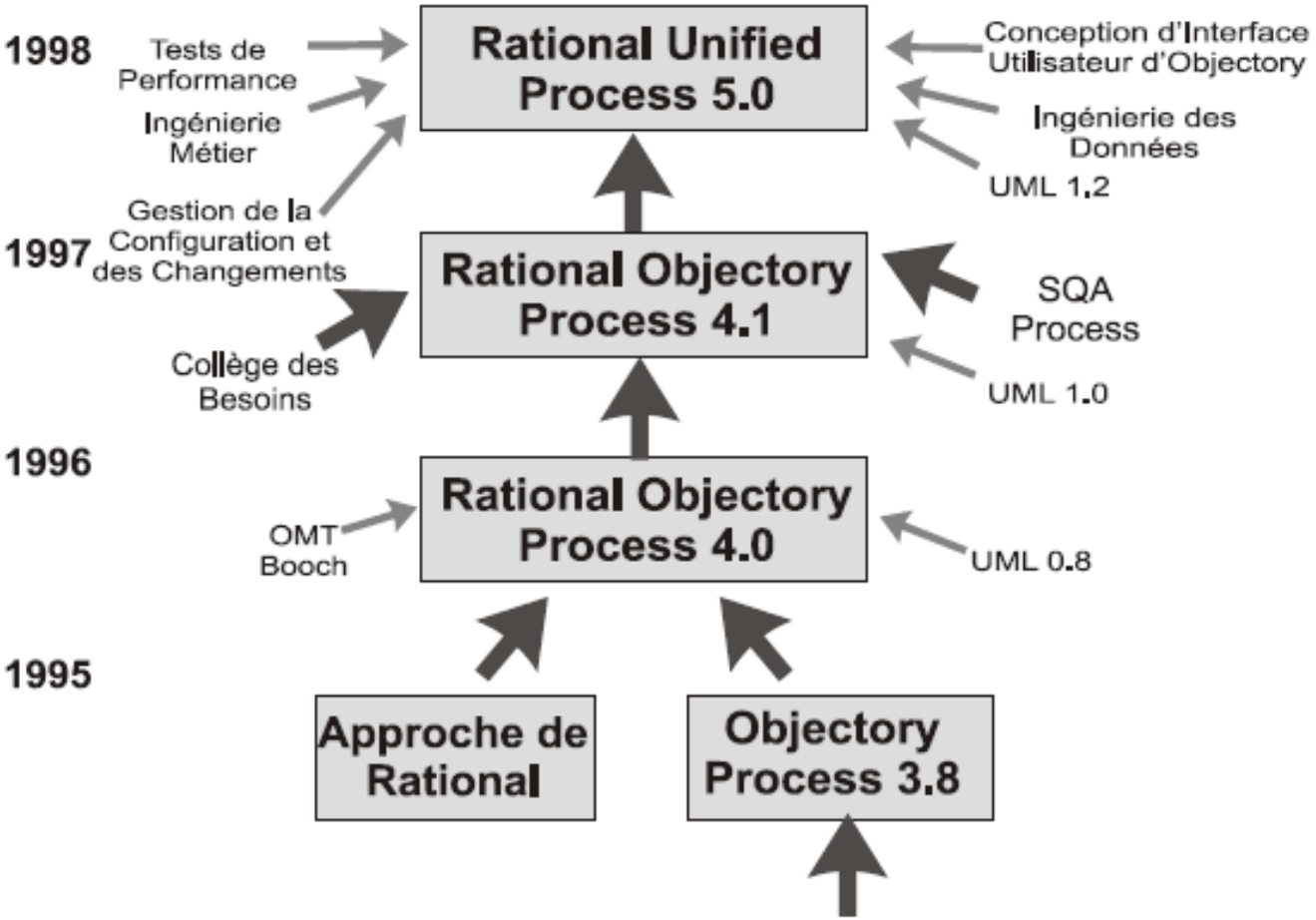
- Architecture: The fundamental concepts or properties of a system in its environment including these elements, relationships and the principles of their designs and evolutions

- Pattern: describes both a problem that occurs very frequently in the environment and the architecture of the solution to this problem in such a way that one can use this solution thousands of times without ever adapting it in the same way twice.

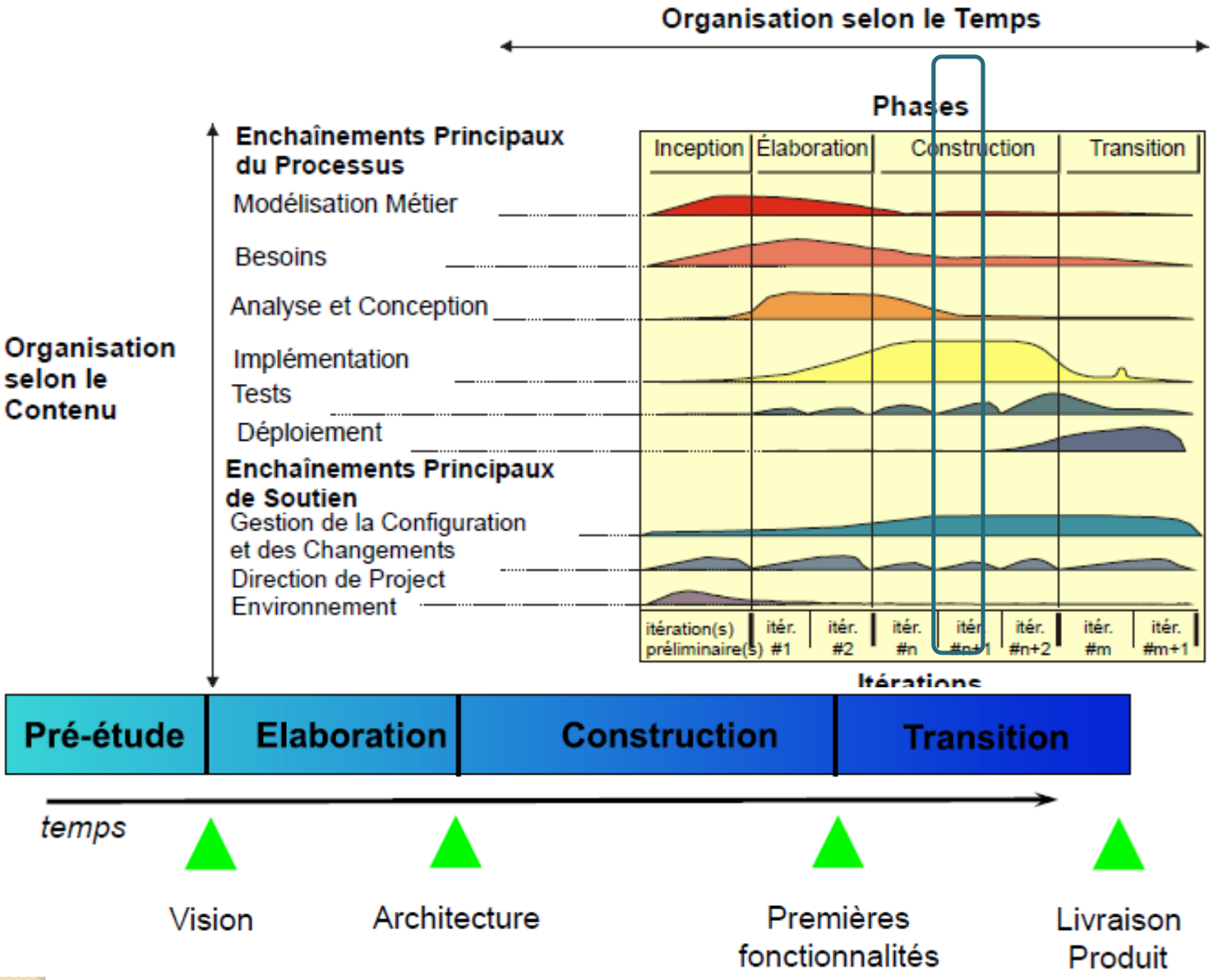
The Rational Unified Process (RUP)

- The Rational Unified Process (RUP)
  - is one of the most famous implementations of the PU method
- Characteristics of the RUP:
  - Iterative
  - Model usage (UML)
  - Centered on architecture Driven by use cases
  - Supports object-oriented techniques
  - Configurable process
  - Quality control and risk management

RUP History


1998
Tests de Performance
Ingénierie Métier
Gestion de la Configuration et des Changements
1997
Collège des Besoins
1996
OMT Booch
1995
Approche de Rational
Objectory Process 3.8
Rational Objectory Process 4.0
Rational Objectory Process 4.1
Rational Unified Process 5.0
UML 0.8
SQA Process
UML 1.0
Conception d'Interface Utilisateur d'Objectory
Ingénierie des Données
UML 1.2

RUP life cycle



Definition

- Phase: period of time between major (important) milestones in the development process and during which a set of objectives is achieved A
- milestone: a set of artifacts that have reached a previously set achievement (landmark)
- There are two types:
  - A major milestone: each phase ends with a major milestone, it is a milestone where important commercial decisions are made,
  - A minor milestone: intermediate milestone between two major milestones
- example: end of iteration milestones
- Iteration: represents a complete development cycle

Artifacts

- Any work product It is the specification of physical information used or produced by a software development process
- An artifact can be:
  - A model
  - A template element
  - A document
  - Source code
  - Executables
  - ……..

Inception (Pre-study)

- Delimit the scope of the system,
- Define boundaries and identify interfaces
- Develop use cases
- Describe and sketch the candidate architecture
- Identify the most serious risks
- Demonstrate that the proposed system is able to solve the problems or support the objectives set

Vision: Glossary, Determination of stakeholders and users, Determination of their needs, Functional and non-functional needs, Design constraints

Elaboration

- Specification of the foundations of the architecture, create a reference architecture
- Identify risks, those likely to disrupt the plan, cost and schedule,
- Define the quality levels to be achieved,
- Formulate use cases to cover approximately 80% of the functional needs and plan the construction phase,
- Project planning, developing a bid addressing schedule, personnel and budget issues

- **Architecture**: Software architecture document, Different views depending on the stakeholder, Behavior and design of system components

Construction

- Extension of the identification, description and realization of use cases
- Finalization of analysis, design, implementation and testing
- Preservation of architectural integrity
- Monitoring of critical and significant risks identified in both early phases and risk reduction
- **Product**

Transition

- Preparation of activities
- Recommendations to the customer on updating the software environment
- Development of manuals and documentation concerning the version of the product
- Software adaptation
- Correction of test-related anomalies
- **Latest fixes Product delivery to users**

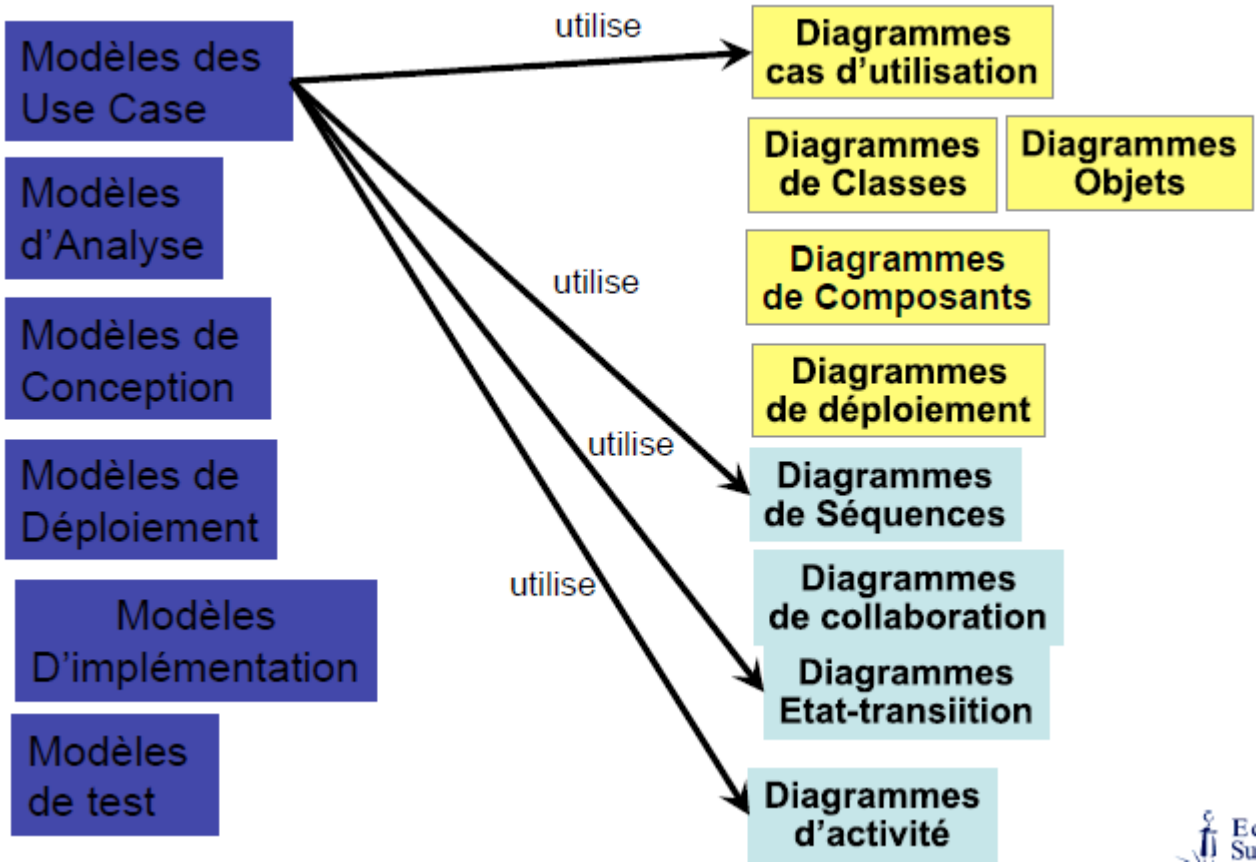
Workflow

- Business modeling:
  - Understanding the structure and dynamics of the organization
  - Understand the problems posed in the context of the organization
  - Design of a glossary
- Collection and expression of needs:
  - With clients and project stakeholders
  - What the system should do
  - Expression of needs in use cases
  - Specifications of use cases in scenarios
  - Functional limits of the project
  - Non-functional specifications
  - Planning and cost forecasting
  - Production of HMI models

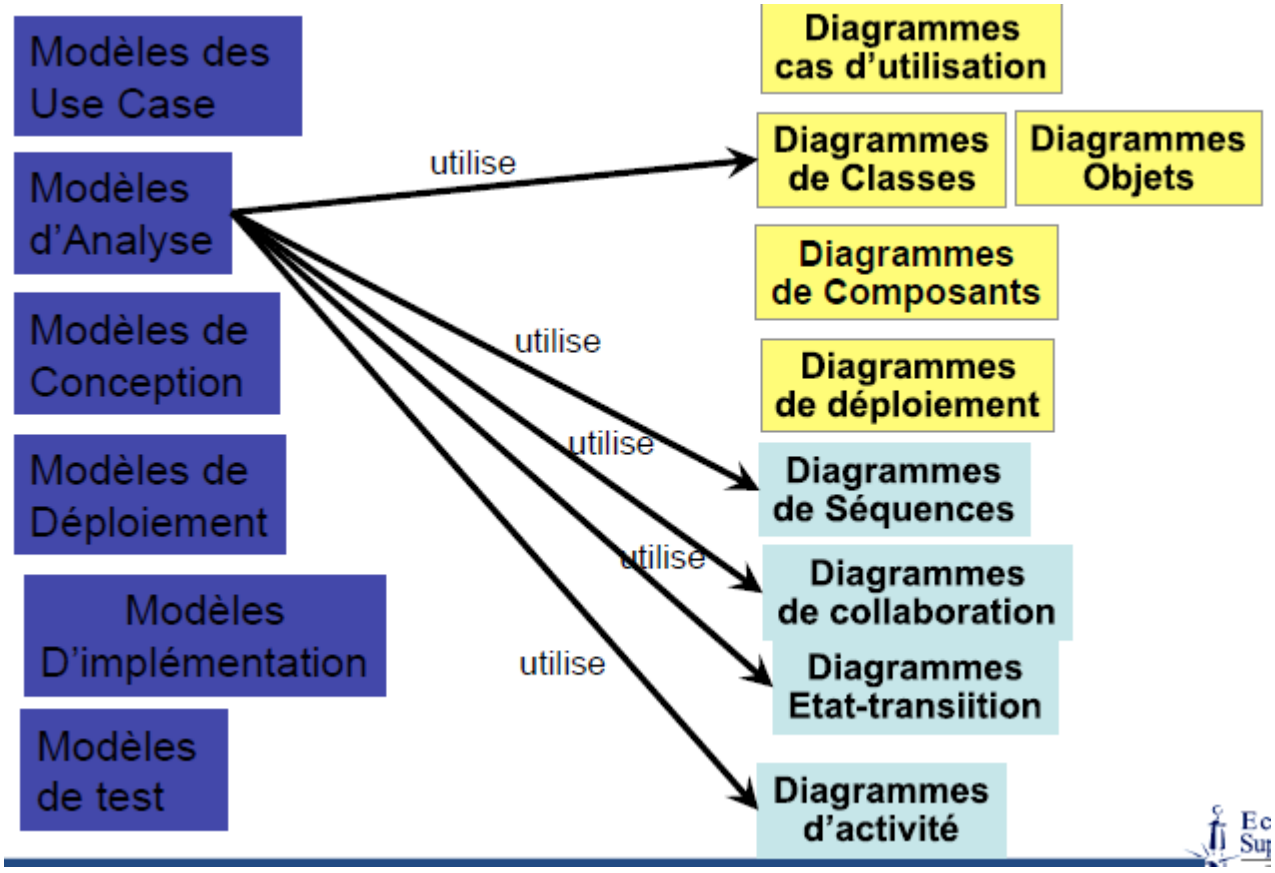
Workflow

- Analysis and design:
  – Transformation of user needs into UML models
  – Analysis model and design model
- Implementation:
  – Iterative development
  – Layering
  – Architectural components
  – Retouching models and needs
  – Independent units, different teams
- Testing, Deployment, Configuration and management of changes, Project management, Environment, Deployment

Using diagrams



Using diagrams



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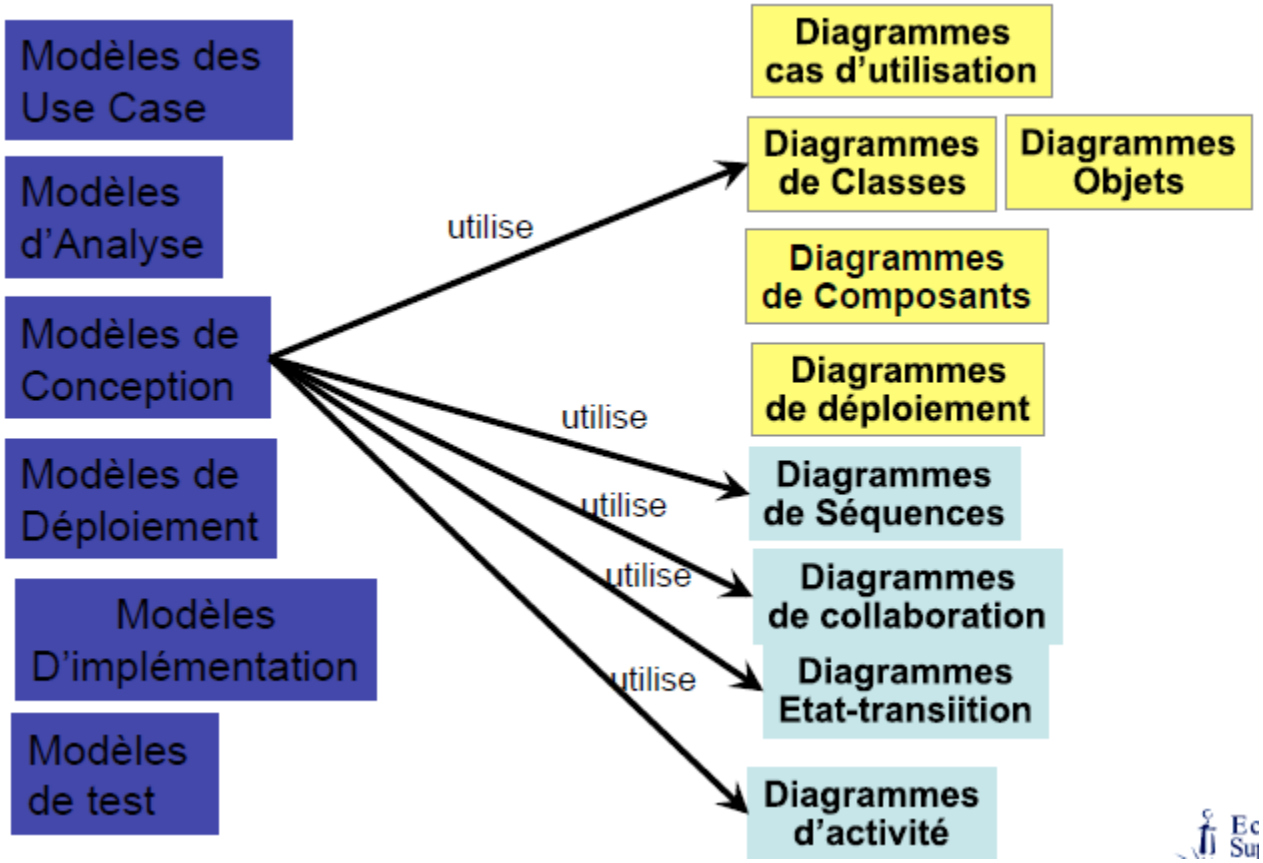

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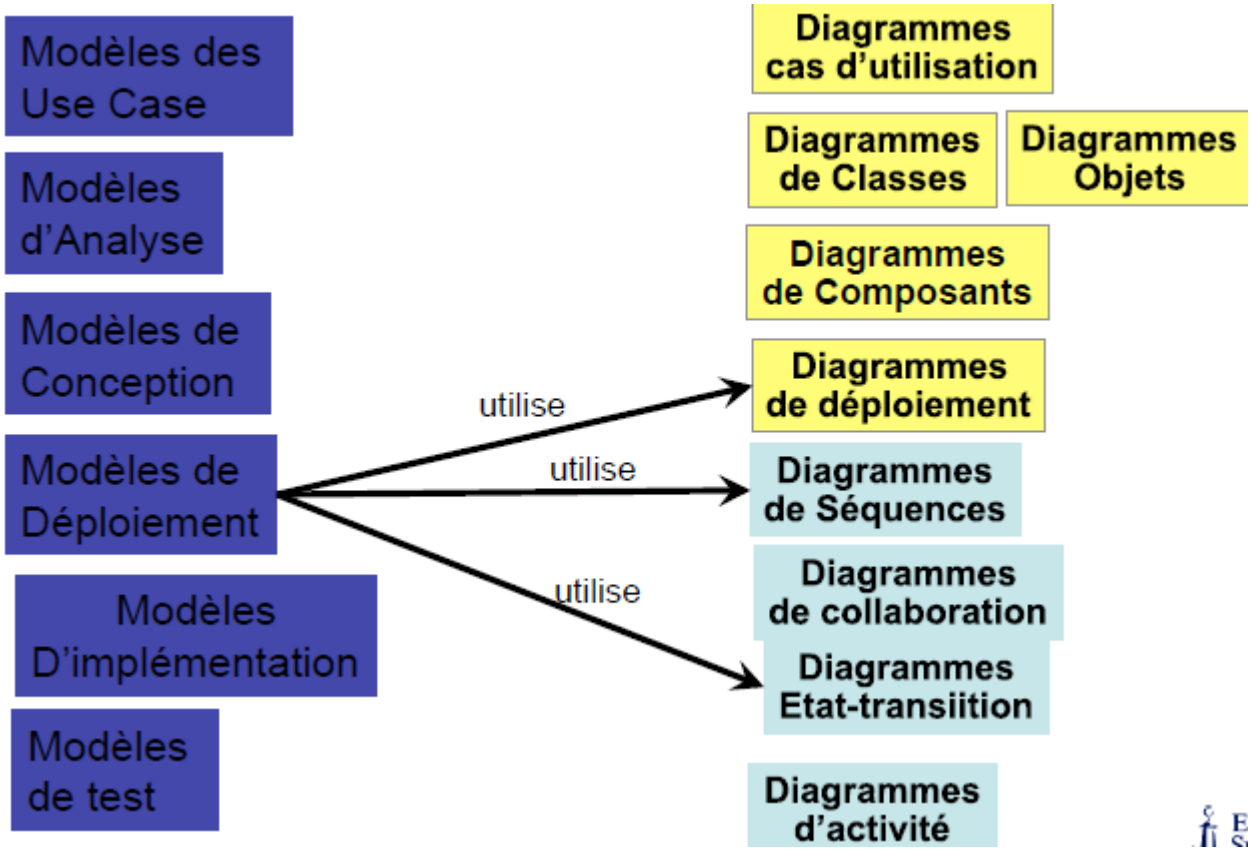

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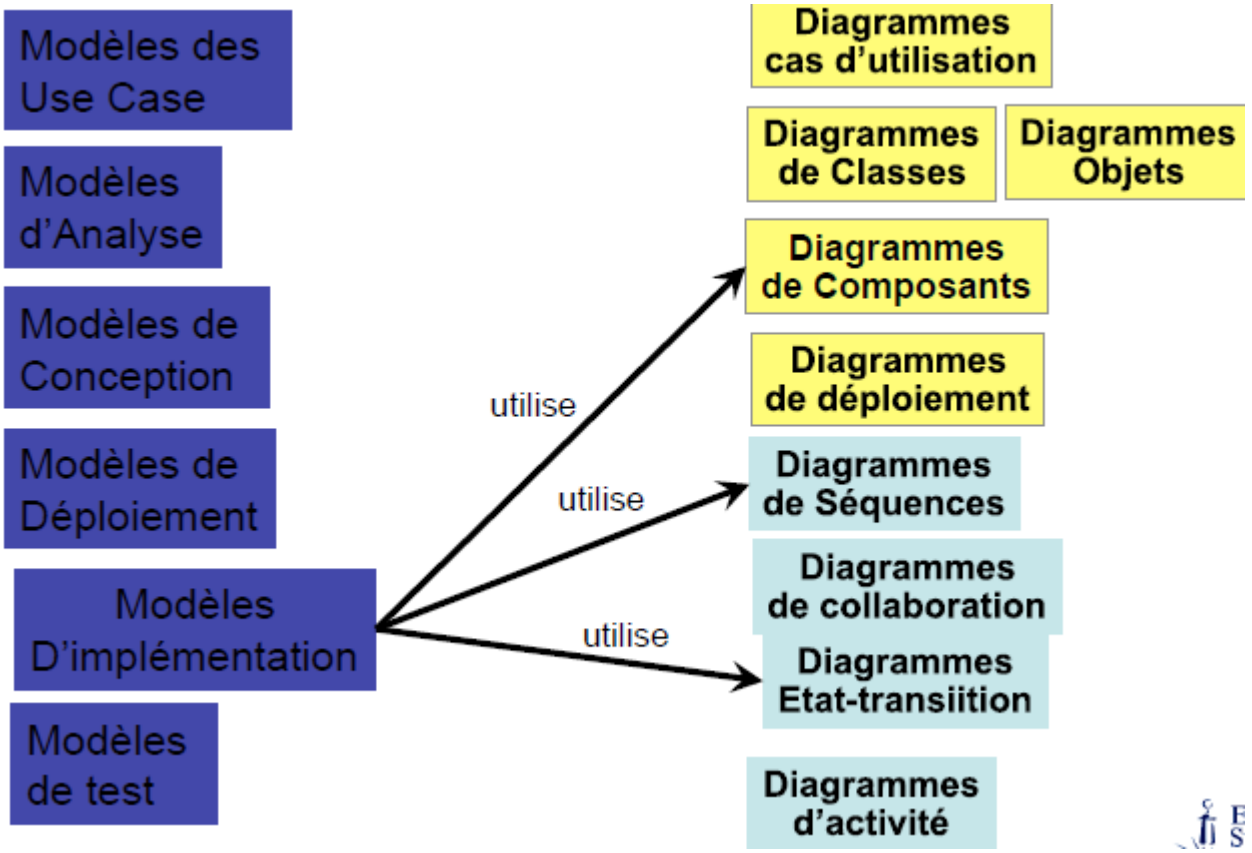

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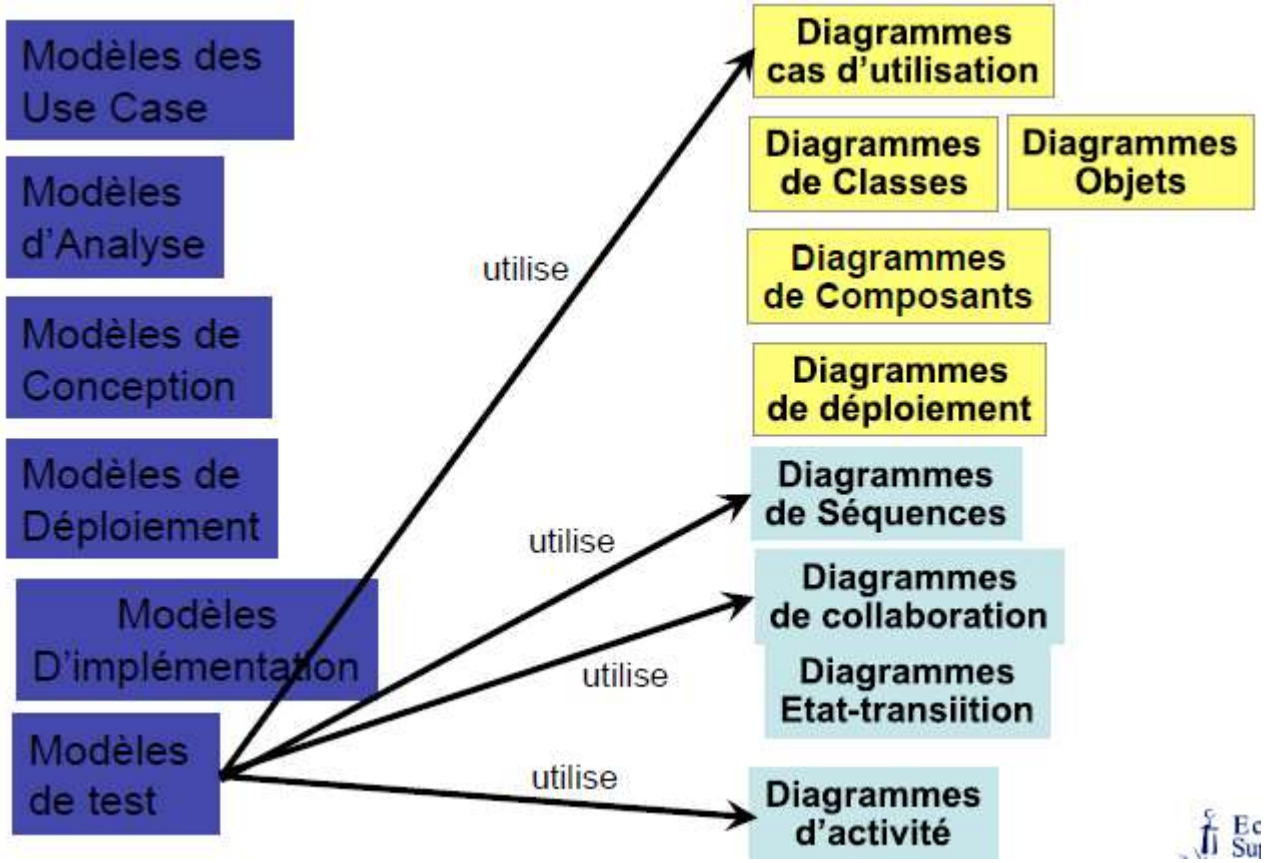

Using diagrams